

Optimal contract policies for improving access to child and youth public funding: the case of the Ontario Autism Program

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Abstract: In 2024, waiting times for autism intervention funding under the Ontario Autism Program ranged from 5 to 7 years, reflecting broader global challenges in accessing autism services. These delays burden families with late interventions and high out-of-pocket costs, impacting both child development and family well-being. This paper develops a contract-based queueing game model between families and an autism funding provider to yield insights on how to improve access to public funding and intervention services. Families choose between waiting exclusively for public funds or seeking private care while waiting, with decisions following a threshold policy based on sensitivity to delays. The funding provider sets the optimal service rate for assessments and subsequent funding allocation, and may also implement a rebate scheme to balance quality and efficiency. We first analyze a general setting of the problem with retrospective rebates which partially offset private expenses, and then study a no retrospective rebate model commonly used in practice. We then extend our framework to an integrated care system as the benchmark, in which a central planner coordinates incentives to optimize overall system efficiency and maximize social welfare. Analytical and numerical results show that calibrated rebates can increase service capacity, thus reduce waiting times, making the system closer to the integrated model. These insights inform policy design for contract mechanisms that enhance autism funding access, with broader implications for healthcare systems worldwide.

Key words: Healthcare operations, Autism spectrum disorder, Queueing game theory, Patient behaviour, Rebate schemes

1 Introduction

Access to publicly funded services in Canada has declined significantly, with the Ontario Autism Program (OAP) serving as a key example. Designed to support children with autism spectrum disorder (ASD), and their families, this program has over 54,000 registered children as of June 2022, many of whom face waiting

times of up to seven years before receiving provincial funding for essential services and treatments (Jones 2022). The challenges extend beyond Canada, as many other high-income countries like the United States and the United Kingdom struggle with intervention systems, making it difficult for caregivers to access timely and coordinated services (Young et al. 2019).

Timely intervention is critical for improving long-term outcomes, yet prolonged waiting times for autism-related services continue to restrict access to these critical services, leading to adverse developmental and health outcomes (Walsh et al. 2021). Empirical studies also show that early interventions lead to long-term positive outcomes for autistic children and their families (Fein et al. 2016, Babalola et al. 2024). In Ontario, Piccininni et al. (2017) highlight how the gap between intensive behavioral intervention (IBI) service demand and supply impacts outcomes, stressing that timely access improves independence and reduces long-term costs from both provincial and societal perspectives. Similarly, Makrygianni et al. (2018) demonstrates the significant benefits of applied behavioural analysis (ABA) programs in enhancing cognitive and communication skills in autistic children.

In addition to the importance of early intervention, families face substantial financial strain due to lengthy waiting times for funding access and the need for private alternatives. Some intensive intervention programs, such as ABA, can cost up to \$90,000 per year, making private alternatives unaffordable for many (Jones 2022). These financial burdens are compounded by the emotional toll of navigating a complex system with uncertain timelines, further complicating efforts to secure timely care.

Given these barriers, there is an urgent need for coordinated, equitable policy solutions. Shepherd and Waddell (2015) advocate for coordinated autism policies that ensure equitable access to services across regions, prioritize resource allocation, provide financial support for families, and train healthcare providers to meet the diverse needs of autistic individuals.

Before presenting the model, we briefly outline how the OAP operates. The ASD diagnosis typically involves assessments by a registered psychologist, pediatrician, and speech-language pathologist, with input from parents and educators. The psychologist compiles a diagnostic report recommending services such as ABA, speech-language pathology (SLP), and occupational therapy (OT). Families use this report to apply for the OAP funding but are placed on a waitlist. When eligible, they attend a single online Determination of Needs (DON) meeting which lasts up to four hours via the AccessOAP platform. During this meeting, a care coordinator assesses child's needs across ten domains. Care coordinators typically have a combination of relevant educational backgrounds (e.g., behavior therapy or early childhood education) and substantial experience in autism and developmental services gained through extensive training. Funding decisions follow within two weeks and can be managed through lump-sum or reimbursement options to cover eligible services. The shortage of behavioural service providers in Ontario has been attributed to high educational and training requirements, emotional demands of the role, and relatively low compensation, all of which

contribute to recruitment challenges and high turnover (Ontario Ministry of Children, Community and Social Services 2021).

In this paper, we develop a novel contract-based queueing game model that provide insights on how to improve access to public funding and timely care for families with autistic children. The model captures the interactions between families seeking funding for autism-related services and an autism funding provider (AFP) responsible for allocating financial support based on each child's needs. Unlike existing queueing models, our framework uniquely focuses on the strategic behavior of families who must decide whether to pursue private care while awaiting an assessment required for accessing public funding—shifting the decision point from joining the queue to how to act while waiting. Additionally, the model accounts for families' sensitivity to both access and quality of care, explicitly capturing how these factors jointly influence their decision-making.

We consider a setting in which families of children diagnosed with ASD, exhibiting heterogeneous delay sensitivities, arrive at the system and join a queue to go through an assessment by the AFP. During the assessment, parents or caregivers undergo an extensive interview regarding their children's needs. After that, funding is allocated based on those needs. Since long waiting times for funding access are the primary barrier to timely intervention, we focus solely on the waiting period before the assessment and do not model the funding amount allocation decision in this paper.

Families face a critical trade-off: they must decide whether to seek out-of-pocket services while waiting to ensure timely intervention and receive the associated monetary lifetime reward, or wait exclusively for the AFP assessment without incurring private costs but delaying the reward from timely intervention. We refer to it as a lifetime reward because it represents long-term public funding until the child turns 18, with earlier access yielding greater benefits. We consider the possibility of a policy in which the AFP offers retrospective rebates to families who choose to seek private services during the waiting period. The AFP also faces its own trade-off between increasing service capacity rate to reduce queue management costs and the financial burden of funding additional capacity. We assume that reducing assessment time may compromise accuracy, leading to misidentification of needs and misallocation of funds. Throughout the paper, when the term service(s) is used in reference to the AFP, it refers to assessment (interview) services for identifying children's needs, and not to the provision of treatment services.

The above trade-offs and observations motivate us to examine the following research questions: (1) How does incorporating the families' lifetime reward and their sensitivity to waiting time into a queueing game framework explain current access barriers in autism services like the OAP? (2) How does implementing retrospective rebates affect outcomes including system waiting time, families' total utility, and AFP's payoff? (3) How do different cost structure parameters and rebate rates influence the optimal decisions of families and the AFP, as well as the resulting outcomes for each stakeholder?

To address these questions, we establish a theoretical model in which the AFP acts as a leader, setting the service capacity rate for assessments to maximize its expected payoff, given the monetary lifetime reward of timely interventions for families, the cost of managing the queue, and the operational costs of running the system. Given the AFP's chosen service capacity rate, families then decide whether to seek out-of-pocket services while waiting. Their objective is to maximize their utility, balancing the monetary reward from timely intervention against the cost of private care. The dynamics of service delivery are modelled using a single-server queueing system with exponentially distributed service times and inter-arrival times.

We first develop a general framework assuming that the AFP provides retrospective rebates to families, partially reimbursing them for out-of-pocket expenses incurred during the waiting period. We then analyze a no retrospective rebate model, which mirrors current practices, such as those in the OAP.

Under the general setting, we show that a family's decision follows a threshold policy, reflecting their sensitivity to funding access delays. We then characterize the limits on the AFP's service capacity rate which regulate the families' decision on whether to seek out-of-pocket options. Interestingly, we find that when the service capacity rate is extremely low but still within steady-state conditions, families are more inclined to wait exclusively for AFP, as out-of-pocket expenses increase with longer waiting times. For the no retrospective rebate model, we analytically characterize the optimal service rate of the AFP and the corresponding optimal decisions for families.

We further extend our analysis to an integrated care system with no rebates, where a central planner optimizes decisions collectively to maximize social welfare, incorporating both families' total utility and the AFP's expected payoff. We analytically derive central planner's optimal decisions and prove monotonicity properties related to families' waiting delay sensitivity thresholds.

Through numerical experiments, we demonstrate that the integrated care system results in a higher service capacity, leading to lower average waiting times, which benefits families. As a result, we use this system as a benchmark. We then show that introducing retrospective rebates can help bridge the gap between integrated care and current no rebate practices by reducing waiting times. However, while rebates improve families' experience, they impose significant financial burdens on the AFP. To address this, we explore strategies for optimally balancing rebate values, ensuring that both families' total utility and the AFP's expected payoff are considered. Further numerical examples illustrate the impact of various model parameters on outcomes for different stakeholders when the balanced rebate values are incorporated.

The organization of the paper is as follows: Section 2 reviews the relevant literature and highlights our contributions. The model is formulated in section 3, followed by an analysis of the general case with retrospective rebates and the baseline case without rebates in section 4. Section 5 develops an integrated care system where a central planner makes key decisions. In section 6, we conduct a sensitivity analysis of the optimal solution, and section 7 presents the implications for different stakeholders. Finally, section 8 summarizes the findings and outlines directions for future research.

2 Literature Review

This paper contributes to a growing stream of operations management research that leverages stylized queueing models embedded in game-theoretic frameworks to examine policy issues in healthcare. This stream focuses on strategic level problems such as capacity and financing decisions and study incentives in queueing systems, extending traditional models that often overlook customer behavior. Naor (1969) was the first to study an observable queue with rational customers who decide whether to join or balk. The model assumes an $M/M/1$ server and homogeneous customers who value service at a certain level while incurring waiting costs per unit of time spent in the system. The results show that a pure equilibrium strategy exists, defined by a threshold value that distinguishes between queue lengths that prescribe joining and those that prescribe balking. This work was later extended to unobservable queues by Edelson and Hilderbrand (1975). For a comprehensive review of efforts integrating economic considerations into queueing theory, see Hassin and Haviv (2003) and, more recently, Hassin (2016).

Our setting, however, diverges significantly from the existing literature in its focus on customers decisions. Unlike prior studies where customers decide either to join a queue or balk and pursue an outside option (e.g., Debo and Veeraraghavan 2014, Dai et al. 2017, Savva et al. 2019), or to choose between private and public providers (e.g., Hua et al. 2016, Zhang and Yin 2021, Zhou et al. 2023), we assume that all customers join the queue because they are unable to forego the financial benefit provided by the AFP once the assessment is complete. Instead, patients may opt for out-of-pocket services while waiting, motivated by the desire for early intervention. Beyond this key distinction, our model captures two critical features not jointly addressed in prior research. First is the trade-off between quality and speed, where longer service times improve quality, reduce risk, and allow for greater customization at the cost of increased congestion and delays (Anand et al. 2011, Kostami and Rajagopalan 2014, Dai et al. 2017). Second is the patient sensitivity to both quality and access to care, following Jiang et al. (2020), explicitly differentiating these factors in their utility by defining a quality-driven reward from timely intervention as well as waiting time in the system.

Our work also intersects with the broader literature on managing the trade-off between affordability and timely access in healthcare systems (e.g., Andritsos and Aflaki 2015, Wan and Wang 2017). For instance, Andritsos and Tang (2014) examine a system where a funder coordinates one public and one private service provider, determining the price per service and server capacities. The private provider charges additional fees for services not covered by public funding, while the funder reimburses treatments performed at the public hospital. Their findings indicate that introducing a private provider reduces waiting times but may increase the funder's costs, depending on the coordination mechanism employed.

In a similar vein, our model captures the financial and timing trade-offs faced by families who must choose between paying out-of-pocket for private services during prolonged wait periods to secure timely

intervention, or relying solely on delayed public funding and postpone care. Unlike models that involve provider competition, we assume all families join a centralized queue managed by a public autism funding provider (AFP). Those seeking earlier intervention via private care incur expenses while continuing to wait for public assessment and funding.

Our problem relates to models incorporating rebates for healthcare expenses. Qian et al. (2017) examine a system with a service manager, patients, and hybrid public-private providers. Private care offers immediate service at a cost, while public care is free but congested (with observable delays). Patients choose based on expected waiting times, individual-specific waiting costs, and private care expenses. A subsidy is provided for private care if public waiting times exceed a threshold, with the service manager determining subsidy amounts and thresholds. Qian et al. (2017) compare two subsidy schemes, identifying conditions where one outperforms the other based on patient costs and serviceability. Guo et al. (2019) further studies the impact of two reimbursement schemes, fee-for-service and bundled payment, on the social welfare, the patients revisit rate, and the patient waiting time in a public healthcare system. In the fee-for-service scheme, the provider receives the payment each time a patient visits, whereas in the bundled scheme, the provider receives a lump sum for the entire episode of care regardless of the patients number of visits. The authors formulate a three-stage Stackelberg game to determine the patients' initial visit rate, the service provider's service rate, and the funder's reimbursement rate. Guo et al. (2019) show that when the patient pool is large, the bundled payment scheme outperforms the fee-for-service model in terms of social welfare and revisit rate, though the fee-for-service model results in shorter waiting times. However, when the patient pool is small, the bundled payment scheme consistently outperforms fee-for-service across all three performance metrics.

Our work aligns with Qian et al. (2017) in that we also assume the government (AFP) provides direct rebates to families rather than to healthcare providers. However, it differs in how family decision-making and rebate mechanisms are modelled. In their framework, patients are directed to private care once their waiting time exceeds a threshold and receive a corresponding rebate. In contrast, our model allows families to independently decide whether to pursue out-of-pocket services, with those who do receiving a retrospective rebate covering a portion of their total spending. The rebate rate is initially treated as an exogenous parameter, and we subsequently determine its balanced value numerically—one that aligns the incentives of families and the AFP.

This study is further related to models examining social optimality, often achieved through the coordination of a central planner (e.g., a government entity) in service systems (e.g., Ghamat et al. 2024) or a centralized supply chain in manufacturing and retail operations (e.g., Adida and DeMiguel 2011), with the goal of maximizing overall system benefits. Using a queueing game framework, Jiang et al. (2020) study a healthcare market where a payer implements a performance-based compensation contract for two competing hospitals. The authors incorporate information asymmetry between the payer and the hospitals

regarding hospital operating costs. The study introduces a bonus compensation mechanism, rewarding each hospital based on the patient benefits it delivers. Assuming the payer exhibits the same level of altruism as a social planner, the authors demonstrate that both heightened competition among hospitals and the inclusion of bonus incentives can significantly improve patient outcomes.

Our work diverges from existing literature in its treatment of the social planner's role. Specifically, the double marginalization phenomenon commonly observed in supply chains is not applicable in our context because there are no monetary transfers between the AFP and families prior to fund allocation, setting it apart from conventional frameworks such as Chen and Tang (2022) and Li et al. (2022).

3 The Model

We analyze a system comprising an autism funding provider (AFP) and families with autistic children. Our model captures the trade-offs faced by these families. Specifically, they are motivated to seek out-of-pocket services while waiting for their AFP assessment in order to ensure timely intervention for their child. As reviewed in the introduction, extensive research highlights that early intervention leads to better outcomes, enhancing the overall benefit to families. However, this comes at a significant financial cost, as families must cover out-of-pocket expenses for any chosen service provider (SP). Families opting for private care incur out-of-pocket expenses paid directly to their chosen SP.

To examine these dynamics, we first introduce a general framework where the AFP offers retrospective rebates, reimbursing a portion of the costs incurred by families during the waiting time in the form of a rebate. We then discuss a special case in which the AFP does not offer any reimbursement, mirroring the current situation in systems such as the Ontario autism program (OAP).

The AFP trade-off lies between expanding service capacity to reduce queue management costs plus the rebate costs, and the financial burden of funding this additional capacity as modelled in Mendelson (1985). Additionally, increasing the service capacity rate (given a fixed number of servers due to workforce shortages) may reduce time per assessment, potentially compromising the accuracy and quality of assessment (Alizamir et al. 2013).

3.1 Families' Decision Problem

We assume that families with autistic children arrive at the system according to a Poisson process with rate λ and join the queue to be assessed for funding allocation by the AFP with rate μ . We model the service delivery dynamics during the AFP's assessment as an $M/M/1$ queue. While an $M/M/s$ queue would more accurately represent multiple assessors, we adopt an $M/M/1$ approximation to enhance analytical tractability. This simplification facilitates the examination of how waiting time influences families' strategic decisions and informs the design of funding incentives. Such approximations are common in the queueing literature when the goal is to derive managerial insights rather than compute exact performance metrics (e.g.,

Naor 1969; Hassin and Haviv 2003). Let $w_s(\lambda, \mu)$ be the corresponding waiting time function evaluating the mean time patients spend in the system, i.e., waiting in the queue and/or being served, with $\mu > \lambda$.

We assume families are heterogeneous in their sensitivity to waiting time, which is represented by a random variable v that follows a uniform distribution between zero and one, i.e., $v \sim U[0, 1]$. A value of v closer to one indicates more sensitivity towards waiting time. Consistent with the modelling approach of Liu et al. (2023), we define total out-of-pocket expenses as the sum of a base fee and a variable cost, differing in that we incorporate heterogeneity in delay sensitivity. Upon arrival, each family observes the expected waiting time and joins the queue. Some may decide to seek out-of-pocket services while waiting, incurring a base fee of H , which reflects the monetary effort required to find and secure a service provider. Those seeking out-of-pocket services further incur a variable cost of P for each unit of time spent in the system, i.e., the waiting time to receive funding from the AFP plus the assessment duration. We assume that families have accurate knowledge of expected waiting times when they make the decision on whether to seek out-of-pocket services. This knowledge is gained through experiences with the AFP, recommendations from others, or digital platforms that report typical AFP waiting times.

We assume the AFP offers a retrospective rebate to families who seek out-of-pocket services, as a way to partially offset their monetary expenses incurred while waiting for funding. A similar rebate mechanism has been used for Medicare Advantage plans in the form of supporting supplemental benefits (Nicholas and Wu 2020). Let $\gamma \in [0, 1]$ be the rebate rate for the total out-of-pocket expenses.

Patients receive a lifetime, quality-driven reward $R(\mu)$ in monetary terms for timely intervention. We assume that families who choose to seek out-of-pocket services while waiting receive this reward immediately as they begin services right away; therefore, their reward is not subject to discounting. In contrast, families who opt to wait exclusively for the AFP receive the reward only after assessment completion. As such, their expected lifetime reward is discounted based on their waiting time in the system—the longer the wait, the lower the reward. This reflects that delayed access to services diminishes the benefits of timely intervention and improved outcomes (Singhi and Malhi 2023). Additionally, we assume that families who are more delay-sensitive apply a heavier discount to the reward, reflecting their reduced perceived value of benefits obtained after a delay. For example, a family with urgent developmental concerns may be unwilling to wait several months for public funding, viewing the delayed intervention as significantly less valuable. See figure 1 for a schematic illustration of the event sequence.

We further assume that the lifetime reward increases as the service rate decreases (given a fixed one-server setting), because a slower service rate allows care coordinators to spend more time with parents, resulting in more accurate assessments and more precise identification of children's needs. Following Dai et al. (2017), who examines a medical imaging test-ordering setting, we model the lifetime reward as a function of service rate μ as

$$R(\mu) := R_b + \alpha(\mu_b - \mu)^+, \quad (1)$$

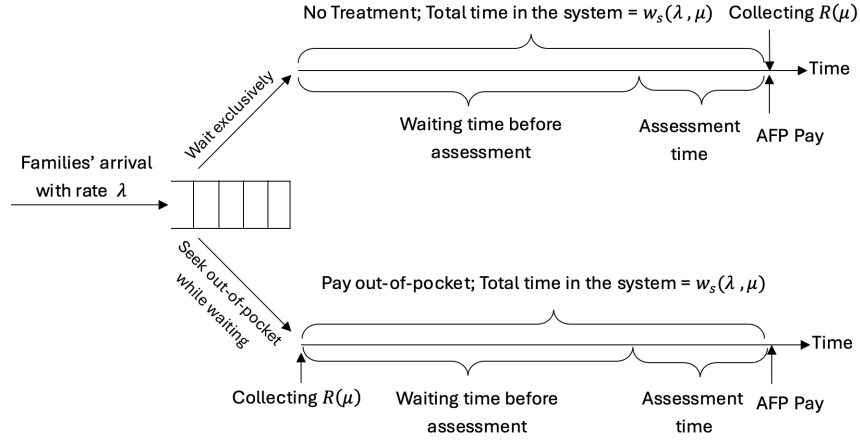


Figure 1 A schematic illustration of the sequence of events for families.

where $x^+ = \max(0, x)$; R_b is the baseline lifetime reward for patients; μ_b refers to the baseline service rate; that is $R(\mu_b) = R_b$; α represents the rate at which the reward increases as service rate decreases. We define μ_b as the maximum feasible service rate at which the AFP can process each application due to bureaucratic procedures or clinical guidelines. This implies that there is a minimum required time to process each application.

Assumption 1 We assume $\mu < \mu_b$ in equation (1), ensuring that the lifetime reward is not constant.

Under this assumption, the reward function simplifies to $R(\mu) := R_b + \alpha(\mu_b - \mu)$. Given the AFP's service rate μ , a family considers seeking out-of-pocket services only if the net reward exceeds the costs, i.e., if $R(\mu) > (1 - \gamma)(H + Pw_s(\lambda, \mu))$, which ensures that the out-of-pocket option is viable. If this condition is satisfied, then the family compares the utility of acting immediately versus waiting exclusively for AFP-funded services. Specifically, they seek out-of-pocket services if

$$R(\mu) - (1 - \gamma)(H + Pw_s(\lambda, \mu)) > \frac{R(\mu)}{1 + \nu w_s(\lambda, \mu)}, \quad (2)$$

where the right-hand side represents the expected discounted reward from waiting, obtained by replacing the moment generating function of w_s with rate $-\nu$, incorporating the families' heterogeneity in their sensitivity to waiting for services. If the initial viability condition is not met, then the family chooses to wait exclusively for the AFP funding. In what follows, we focus on the case where out-of-pocket care is viable, and the decision is governed by inequality (2).

Families with higher delay sensitivity levels are more likely to seek out-of-pocket services to ensure timely intervention. As ν increases, the motivation to opt for out-of-pocket services changes monotonically. As a result, the families' optimal decision follows a threshold policy, as shown in lemma 1.

LEMMA 1. A family's decision on whether to seek out-of-pocket services follows a threshold policy based on their delay sensitivity.

Lemma 1 states that under the optimal service rate policy, families choose out-of-pocket services if and only if their delay sensitivity exceeds a certain threshold, denoted by $v_T(\mu)$. All proofs are provided in the e-companion of the paper, and a glossary of notations is given in appendix §8.

3.2 AFP's Decision Problem

The AFP is responsible for assessing families' needs and, based on this evaluation, allocating funds at later stages. The AFP decides on the service rate μ , which measures the rate of assessment. Let c_w denote the cost of managing the queue per unit of time that patients spend in the system. This captures administrative overhead, coordination efforts, and the strain on limited resources. Consistent with the idea that private care partially relieves pressure on the public system, we assume that families who seek out-of-pocket services require less administrative engagement from the AFP and therefore impose only a fraction of the queue management cost. For example, they may require fewer reminders, follow-ups, or status updates. To reflect this, we introduce the AFP *out-of-pocket care factor*, denoted by $\beta \in [0, 1]$, which scales the effective queue management cost for families who pursue private care options. Further, we denote by $C(\mu)$ the cost per unit of time associated with operating service at rate μ (e.g., employees' wages). We assume that the AFP is altruistic and cares for the well-being of the children, thus aims to maximize the quality-driven reward received by the families (in monetary terms), while minimizing the total expected cost of managing the queue plus operating cost. Therefore, we formulate AFP's problem as follows:

$$\begin{aligned}
 V^*(\mu) = \max_{\mu} & \left\{ \lambda R(\mu) - \lambda \int_{v_T(\mu)}^1 [\gamma(H + Pw_s(\lambda, \mu)) + \beta c_w w_s(\lambda, \mu)] dv \right. \\
 & \left. - \lambda \int_0^{v_T(\mu)} c_w w_s(\lambda, \mu) dv - C(\mu) \right\}, \\
 \text{s.t.} \quad & \mu > \lambda \quad (\text{stability condition}), \\
 & \mu < \mu_b \quad (\text{Assumption 1}), \\
 & v_T(\mu) \quad (\text{as defined in lemma 1}).
 \end{aligned} \tag{3}$$

In problem 3, the AFP determines the optimal service rate while accounting for families' decisions on whether to seek out-of-pocket services. The value function comprises four key components: (1) the lifetime quality-driven reward in monetary terms for all families, (2) the expected rebate and reduced queue cost for families who choose out-of-pocket services, (3) the queue cost for those who wait exclusively for the AFP funding, and (4) the service operation cost. Note that in problem (3), since $v \sim U[0, 1]$, the expected total number of families seeking out-of-pocket and waiting exclusively for AFP services are calculated as $\lambda(1 - F(v_T(\mu))) = \lambda(1 - v_T(\mu))$ and $\lambda F(v_T(\mu)) = \lambda v_T(\mu)$, respectively, where λ is the arrival rate. We denote the optimal solution to problem (3) by μ^* , and the corresponding optimal delay sensitivity threshold in the families' problem by $v_T(\mu^*)$. Figure 2 summarizes the overall decision-making processes and funding flows among the various entities in the model. We begin by focusing on the bottom part of the figure, which illustrates the system dynamics without the role of the central planner.

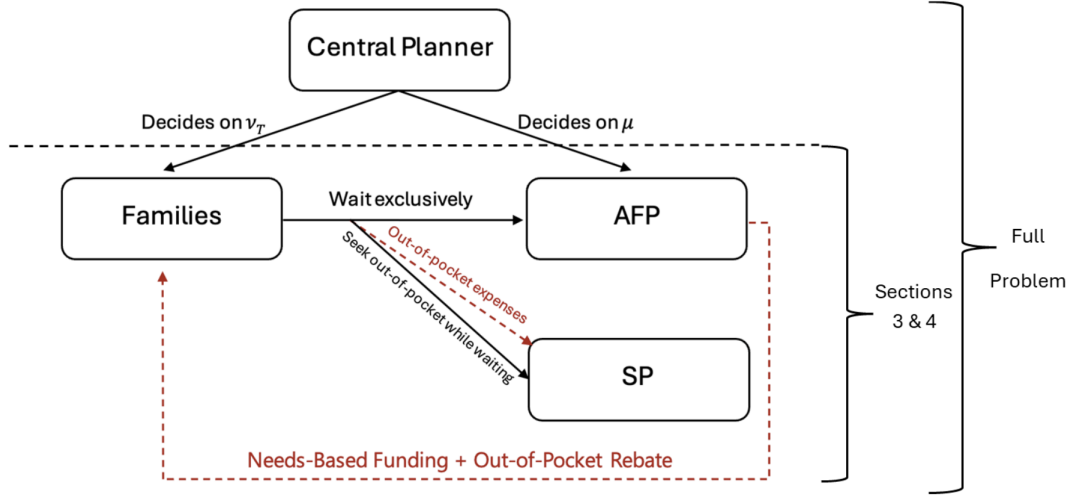


Figure 2 Different players in the system, their decisions (black solid arrows), and the flow of funds (red dashed arrows).

4 Analysis

In solving the problems, we assume that the arrival rate is normalized to one, i.e., $\lambda = 1$. Following Hassin (2007), we assume a linear operation cost $C(\mu) = \kappa\mu$ for some $\kappa > 0$, defined as a multiplier. For mathematical tractability, we normalize key parameters as follows: the rate α at which the reward decreases with service rate, the fixed cost of out-of-pocket services, H , and the coefficient of operation cost, κ . Specifically, we set $\alpha = 1$, $H = 1$, and $\kappa = 1$.

Given lemma 1, a family with delay sensitivity $v_T(\mu)$ or higher will choose out-of-pocket services since the utility of this option exceeds that of waiting exclusively for the AFP services. Specifically, solving the inequality $R(\mu) - (1 - \gamma)(H + Pw_s(\lambda, \mu)) > \frac{R(\mu)}{1 + vw_s(\lambda, \mu)}$ for v , we obtain the delay sensitivity threshold as

$$v_T(\mu) = \frac{(1 - \gamma)(\mu - 1)(\mu + P - 1)}{(\mu - 1)(\gamma + R_b + \mu_b - \mu - 1) - (1 - \gamma)P}. \quad (4)$$

For ease of exposition, we introduce two bounds on μ , denoted l_1 and l_2 , within which the delay sensitivity threshold v_T remains within the feasible domain $(0, 1)$. That is, $v_T \in (0, 1)$ if $l_1 < \mu < l_2$; outside this range, all families either wait or seek out-of-pocket services. See the e-companion for definitions of l_1 and l_2 . These extreme cases do not reflect practical scenarios where some families pursue services while others choose to wait exclusively. We now state:

LEMMA 2. *For any given μ , the optimal decision for families is obtained as*

1. *if $l_1 < \mu < l_2$, then they seek out-of-pocket service for $v > v_T(\mu)$,*
2. *if $\mu \geq l_2$ or $\mu \leq l_1$, then all families wait for AFP services.*

Lemma 2 implies that if the AFP's service rate is within a reasonable range, only families with high delay sensitivity will opt for out-of-pocket services (see the red-shaded region in figure 3). When service is quite

fast, then no families choose to seek out-of-pocket services, as the program's offering will be sufficient to meet their needs (see the blue-shaded region to the right of l_2 in figure 3). Counterintuitively, when the service is extremely slow but still under steady-state conditions ($\mu > \lambda$), families are more inclined to rely exclusively on the AFP. This is because the system becomes so congested, such that the prolonged wait leads to prohibitively high out-of-pocket expenses (see the blue-shaded region to the left of l_1 in figure 3).

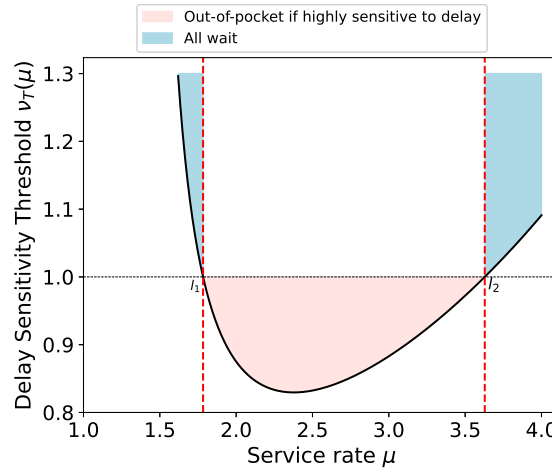


Figure 3 Families' delay sensitivity threshold $v_T(\mu)$ as a function of service capacity rate μ with $P = 5$, $R_b = 5$, $\mu_b = 6$, and $\gamma = 0.3$.

The following corollary outlines the properties of $v_T(\mu)$:

COROLLARY 1. *The delay sensitivity threshold $v_T(\mu)$ in equation (4) is convex in μ (i.e., $\frac{d^2 v_T(\mu)}{d\mu^2} > 0$), increasing in P (i.e., $\frac{dv_T(\mu)}{dP} > 0$), and decreasing in R_b (i.e., $\frac{dv_T(\mu)}{dR_b} < 0$).*

The above corollary implies that as the baseline lifetime reward R_b increases, the delay sensitivity threshold reduces, leading to a larger number of families opting for out-of-pocket services while waiting for AFP assessment. This is because families recognize that they can gain more by starting intervention immediately rather than waiting for AFP funds. Discounting this value also results in a greater loss for them if they choose to wait exclusively. Conversely, as the variable cost for out-of-pocket services P increases, the delay sensitivity threshold increases, leading to more people deciding to wait for AFP. This is because out-of-pocket services become more expensive, making the option of waiting more attractive. Further, as it is shown in figure 3, there exists a threshold in the service rate, μ , below (above) which $v_T(\mu)$ decreases (increases) with μ . This behavior stems from the trade-off that families face: choosing between seeking out-of-pocket services to obtain an immediate lifetime reward at the expense of private costs, or waiting exclusively for the AFP to avoid out-of-pocket expenses while accepting a discounted reward due to the delay. Up to a certain threshold, increasing the service rate reduces waiting time and, consequently, lowers the total private spending, encouraging more families to pursue out-of-pocket care. However, this service

rate is still insufficient to make waiting exclusively the preferred option. Beyond this threshold, the service rate becomes high enough such that families begin to favor waiting for the AFP, as the delay is minimal.

Determining the closed form analytical expression for the optimal service capacity rate, μ^* , in the rebate model is intractable. Therefore, we provide a numerical example to examine the impact of rebates on the optimal service capacity μ^* , and to show when offering rebates may be beneficial. The results, illustrated in figure 4 panel (a), reveal that the relationship between rebates and μ^* is not monotonic and may exhibit convexity under certain scenarios. As detailed in corollary 1, the delay sensitivity threshold $v_T(\mu)$ is a convex function of μ with a critical point below which $v_T(\mu)$ decreases as μ increases. In this region, increasing μ prompts more families to seek out-of-pocket services. Consequently, offering rebates does not enhance system performance further, as families are already inclined toward non-AFP services. In such scenarios, the AFP has little incentive to increase service capacity. Beyond this critical point, however, increasing μ encourages more families to remain on the AFP waitlist. In this region, rebates prove effective and results in increasing the optimal service capacity. This dual behavior underscores the nuanced dynamics of rebates and their impact on service capacity optimization. Panel (a) further shows that μ^* increases with c_w , and its sensitivity to γ is more pronounced when c_w is smaller. Panel (b) of figure 4 illustrates the effect of the rebate rate on the service capacity limits defined in lemma 2. The results indicate that as γ increases, l_1 decreases while l_2 increases; however, the decrease in l_1 is relatively small, whereas l_2 is highly sensitive to changes in γ . This suggests that introducing a rebate may expand the range of μ in which families choose out-of-pocket services when $v > v_T(\mu)$.

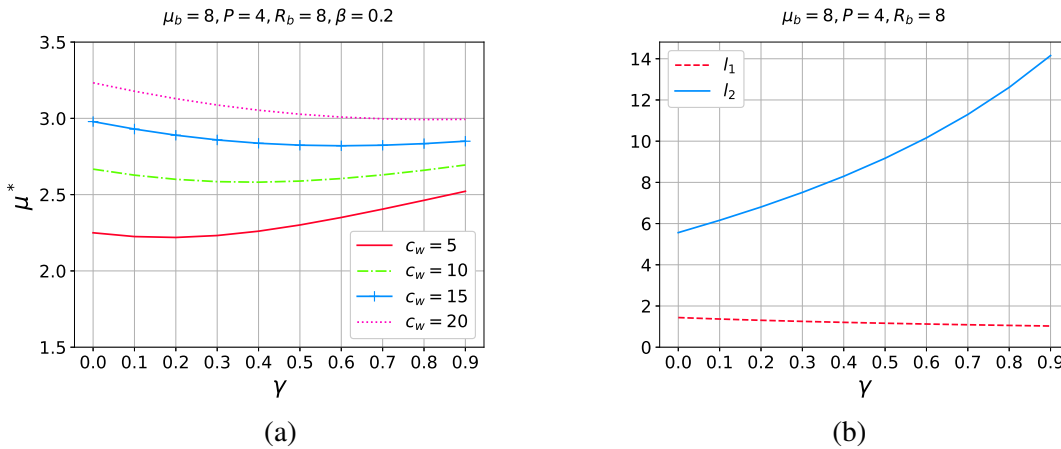


Figure 4 Panel (a): Optimal service capacity, μ^* , as a function of rebate rate γ for different values of c_w ; Panel (b): The limits on μ , l_1 and l_2 , as a function of rebate rate γ .

4.1 Special Case: No Retrospective Rebates

In this section, we consider a special case which represents the current system used in Ontario, assuming that out-of-pocket expenses incurred during the waiting period are not retrospectively reimbursed, i.e., $\gamma = 0$.

Similar results for the delay sensitivity threshold given in equation (4) and families' optimal decision in lemma 2 are derived with $\gamma = 0$. For the rest of the paper, we use superscript n to denote the parameters related to the no retrospective rebate model. The following lemma establishes the relationship between the delay sensitivity thresholds in the models with and without the rebate.

LEMMA 3. *For a given service capacity rate μ , the delay sensitivity threshold in the no retrospective rebate model is higher than in the retrospective rebate model, i.e., $v_T^n(\mu) > v_T(\mu)$.*

The above lemma indicates that offering rebates to families through the AFP incentivizes them to opt for out-of-pocket services more frequently. Consequently, this reduces the delay sensitivity threshold in the rebate model, encouraging broader participation in non-waitlist options.

Knowing the families decision about seeking out-of-pocket services, the AFP decides on its service level solving problem (3) with $\gamma = 0$. Let $\phi(\mu) = \frac{\partial v(\mu)}{\partial \mu}$. We prove in the e-companion that $\phi(\mu)$ is a strictly decreasing function of μ on (l_1^n, l_2^n) , so we can now state how the service rate is chosen.

PROPOSITION 1. *When $\gamma = 0$, the AFP sets the service rate μ such that*

1. *if $\phi(l_1^n) \geq 0$ and $\phi(l_2^n) \geq 0$, then $\mu^* = l_2^n$; all families wait exclusively for the AFP services (i.e., $v_T^n = 1$),*
2. *if $\phi(l_1^n) > 0$ and $\phi(l_2^n) < 0$, then there exists a unique μ^* in the interval (l_1^n, l_2^n) such that $\phi(\mu^*) = 0$; families seek out-of-pocket service for $v > v_T^n(\mu^*)$,*
3. *otherwise, $\mu^* = l_1^n$; all families wait exclusively for the AFP services (i.e., $v_T^n = 1$).*

Proposition 1 shows that the limits for μ^* , l_1^n and l_2^n , provide valuable insights into a family's decision based on optimal service capacity scenarios. In particular, when the service capacity reaches its maximum, l_2^n , (proposition 1.1), all families exclusively opt to wait for the AFP funds. This reflects a *functional system* which efficiently meets demand, removing the need for out-of-pocket expenditures. On the other extreme, at the lowest level of service capacity, l_1^n , (proposition 1.3), all families prefer to wait for the AFP funds. The system becomes so congested that the extended waiting times impose high out-of-pocket costs, deterring families from pursuing private services while they wait. This scenario illustrates the prohibitive impact of financial barriers in a *dysfunctional system*. In scenarios with intermediate service capacity (proposition 1.2), families make decisions based on their delay sensitivity levels. Only families with delay sensitivity above the threshold $v_T^n(\mu^*)$ will opt for out-of-pocket services, while others continue to rely on the AFP funds. This case reflects a *mixed system* which often occurs in practice, for example, the OAP, in Ontario, Canada, where some families wait exclusively for the OAP funds, while others seek private options during their wait depending on their tolerance for delay. These cases underscore how variations in service capacity influence family choices, revealing the delicate balance between affordability, waiting times, and access to care.

Since the impact of rebate on the system is not intuitive as shown in figure 4, we introduce a benchmark scenario where there is a central planner making decisions on everyone's behalf.

5 Integrated Care System

We define an integrated care system in which a central planner makes strategic decisions by accounting for a quality-driven reward aligned with families' expectations—whether discounted or offset by out-of-pocket expenses—unlike the AFP, which myopically assumes a quality-driven reward without discounting it for those who wait exclusively. In particular, central planner's objective is similar to the AFP problem in (3) with zero rebate, except that the quality-driven reward is substituted with the total utility of all families. The central planner chooses the delay sensitivity threshold $v_T^c \in [0, 1]$ and AFP's service rate $\mu^c > \lambda$ simultaneously to maximize overall expected social welfare (see figure 2 for the full structure and dynamics of the problem). We use superscript c to denote the parameters associated with integrated care system. The central planner's objective function is formulated as

$$\Pi^{c*}(v_T, \mu) = \max_{\mu, v_T} \left\{ \lambda \int_0^{v_T} \left(\frac{R(\mu)}{1 + v w_s(\lambda, \mu)} \right) dv + \lambda \int_{v_T}^1 (R(\mu) - H - P w_s(\lambda, \mu)) dv \right. \\ \left. - C(\mu) - \lambda v_T c_w w_s(\lambda, \mu) - \lambda(1 - v_T) \beta c_w w_s(\lambda, \mu) : \mu > \lambda, 0 \leq v_T \leq 1 \right\}. \quad (5)$$

The objective function highlights how the central planner's delay sensitivity threshold and service rate are influenced by the trade-offs between families and the AFP. In particular, the reward of timely intervention motivates families to seek out-of-pocket services (lower v_T). On the other hand, the AFP faces pressure to reduce queue management costs, leading it to prefer a higher service capacity, which increases the number of families opting to wait for AFP funds (higher v_T). Similarly, families prefer shorter waiting times to minimize out-of-pocket expenses when opting for out-of-pocket services and to experience less discounting on rewards when choosing to wait exclusively. Achieving this requires the system to operate at a higher service capacity (larger μ). However, the operational costs and the emphasis on quality-driven rewards incentivize AFP to maintain a system with lower capacity (smaller μ).

Solving problem (5), we obtain the delay sensitivity threshold for the central planner as

$$v_T^c(\mu) = \frac{(\mu - 1)(\mu - 1 + P - (1 - \beta)c_w)}{(1 - \beta)c_w - P + (\mu - 1)(R_b + \mu_b - \mu - 1)}, \quad (6)$$

and the following lemma outlines its properties:

LEMMA 4. *The delay sensitivity threshold in the integrated care system, $v_T^c(\mu)$, is convex in μ (i.e., $\frac{d^2 v_T^c(\mu)}{d\mu^2} > 0$), increases with P (i.e., $\frac{dv_T^c(\mu)}{dP} > 0$) and β (i.e., $\frac{dv_T^c(\mu)}{d\beta} > 0$), and decreases with c_w (i.e., $\frac{dv_T^c(\mu)}{dc_w} < 0$) and R_b (i.e., $\frac{dv_T^c(\mu)}{dR_b} < 0$).*

Lemma 4 indicates that as the cost for managing the queue, c_w , increases, even families with lower delay sensitivity levels tend to switch to out-of-pocket services. Conversely, as the disparity between the queue management cost for those seeking out-of-pocket services and those waiting exclusively diminishes

(β increases), families are more inclined to wait exclusively for the AFP funds. The behaviour of v_T^c with respect to μ , P , and R_b follows a pattern similar to that observed in the model with retrospective rebate, as discussed in section 4.

For ease of exposition, we define two additional values for μ : (i) $l_1^c \equiv 1 + \sqrt{\frac{P+c_w\beta}{2}}$, (ii) l_2^c as the root of equation $\frac{c_w}{(\mu-1)^2} - \frac{\mu_b+R_b}{\mu} + \ln\left(\frac{\mu}{\mu-1}\right)(R_b + \mu_b - 2\mu + 1) = 0$. These points are the optimal service rates μ_c^* for the objective function in problem (5) for the two extreme cases $v_T^c = 0$ and $v_T^c = 1$, respectively. Let $\psi(\mu) = \frac{\partial \Pi^c(v_T, \mu)}{\partial \mu}$. We show in the e-companion that $\psi(\mu)$ is a strictly decreasing function of μ on $(\min(l_1^c, l_2^c), \max(l_1^c, l_2^c))$ under the condition $\zeta < 0$ (see equation (EC.7) in the e-companion). Proposition 2 states how the service rate and delay sensitivity threshold are chosen.

PROPOSITION 2. *Suppose that $\zeta < 0$. The central planner sets the delay sensitivity threshold v_T^c and service capacity rate μ^c such that*

1. *When $l_1^c < l_2^c$:*
 - (a) *if $\psi(l_1^c) \geq 0$, and $\psi(l_2^c) \geq 0$, then $\mu^c = l_2^c$ and $v_T^c = 1$. All families wait exclusively for the AFP services,*
 - (b) *if $\psi(l_1^c) > 0$, and $\psi(l_2^c) < 0$, then there exists a unique μ^c in the interval (l_1^c, l_2^c) such that $\psi(\mu^c) = 0$ and $v_T^c = v_T^c(\mu^c)$,*
 - (c) *otherwise, $\mu^c = l_1^c$ and $v_T^c = 0$. All families seek out-of-pocket services.*
2. *When $l_1^c > l_2^c$:*
 - (a) *if $\psi(l_1^c) \geq 0$, and $\psi(l_2^c) \geq 0$, then $\mu^c = l_1^c$ and $v_T^c = 0$. All families seek out-of-pocket services,*
 - (b) *if $\psi(l_2^c) > 0$, and $\psi(l_1^c) < 0$, then there exists a unique μ^c in the interval (l_2^c, l_1^c) such that $\psi(\mu^c) = 0$ and $v_T^c = v_T^c(\mu^c)$,*
 - (c) *otherwise, $\mu^c = l_2^c$ and $v_T^c = 1$. All families wait exclusively for the AFP services.*

Proposition 2 states that, similar to proposition 1, the optimal service capacity is constrained within a boundary defined by l_1^c and l_2^c . Specifically, when the service capacity reaches its upper limit (propositions 2.1.a and 2.2.c), all families choose to wait exclusively for AFP funds, representing a *functional system*. Conversely, when the service capacity hits its lower limit (propositions 2.1.c and 2.2.a), all families opt for out-of-pocket services, leading to a *dysfunctional system*.

Note that the lower bound of the integrated care system always exceeds that of no retrospective rebate model, as established in corollary 2. This distinction explains why, in the integrated system, families would collectively choose out-of-pocket services when the system is at its capacity lower limit, rather than all opting to wait as they would in a care system with no retrospective rebate. Finally, when the service capacity falls within a intermediary range (propositions 2.1.b and 2.2.b), families with delay sensitivity levels above a certain threshold will opt for out-of-pocket services, resulting in a *mixed system*.

COROLLARY 2. *The lower bound of optimal service capacity in the integrated care system is higher than that of no retrospective rebate model, which in turn is higher than that in the retrospective rebate model, i.e., $l_1^c > l_1^n > l_1$.*

Corollary 3 establishes the relationship between delay sensitivity in the two systems under a fixed value of μ . It implies that at the same level of service capacity, a central planner would direct more patients to out-of-pocket services compared to a system in which families make their own decisions. This is because the central planner simultaneously seeks to reduce the congestion in the AFP system while making allocation decisions on behalf of families.

COROLLARY 3. *For a given μ , the delay sensitivity threshold in the integrated care model is lower than that of no retrospective rebate model, i.e., $v_T^n(\mu) > v_T^c(\mu)$.*

The above relationship, however, does not necessarily hold once their respective optimal values of μ are taken into account since the optimal service capacity is different in the two systems, as presented in corollary 4. This lemma identifies a condition under which the no retrospective rebate model yields a lower optimal service rate capacity compared to the integrated care system. In the integrated system, the central planner may increase the service rate to reduce the waiting times, and thus lowering out-of-pocket expenses. By doing so more families can be directed to out-of-pocket services for immediate reward realization and improved outcomes for autistic children.

COROLLARY 4. *If $\phi(l_1^c) < 0$ and $l_1^c < l_2^c$, then $\mu^{n*} < \mu^{c*}$.*

In the following sections, we explore how introducing retrospective payments can help align the two systems and improve outcomes for both the AFP and families.

6 Numerical Analysis of the Optimal Policies

We present several numerical examples to compare the optimal policies across the three models discussed earlier. Additionally, we evaluate how variations in model parameters affect the optimal service capacity and the optimal delay sensitivity threshold in each of the three systems. To ensure meaningful comparisons, in each system, we focus on scenarios where some families opt for out-of-pocket services while others wait exclusively for the AFP funding (mixed systems), excluding extreme scenarios in which all families make the same choice (functional or dysfunctional systems).

The plots in figure 5 demonstrate that μ^* is higher in the integrated care system compared to the other two models. This is because both the AFP and family benefits are considered in the integrated care system, where central planner balances the service rate μ with the v_T threshold allowing the higher cost of increased service capacity to be offset by greater rewards from directing more families to out-of-pocket services. Introducing a rebate marginally improves the systems' performance by increasing optimal service capacity. This is because the prospect of rebates encourages more families to pursue out-of-pocket services to

receive immediate intervention while increasing the AFP's financial burden. In response, the AFP optimally increases the service capacity to reduce waiting times, making families more willing to wait exclusively for the AFP funds and thus limiting rebate payments. Panels (a) and (b) of figure 5 reveal that the optimal service capacity increases with both the cost of managing the queue, c_w , and the AFP out-of-pocket factor, β . This is because the AFP is incentivized to expand service capacity as the cost of maintaining the queue rises and as its care for families seeking out-of-pocket services increases. Panel (c) illustrates that μ^* increases with out-of-pocket expenses, P . This is attributed to families' preference for waiting for AFP funds when out-of-pocket costs are high, prompting the AFP to enhance its service capacity to accommodate the increased demand.

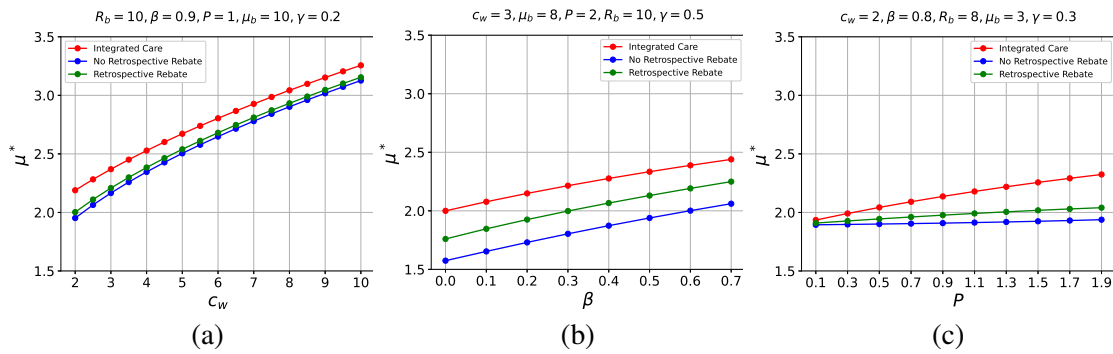


Figure 5 Optimal service capacity, μ^* , as a function of different model parameters.

The plots in figure 6 illustrate that the optimal delay sensitivity threshold is smaller in the model with rebate compared to the no retrospective rebate model, as established in corollary 3. In the integrated care model, v_T^* is also smaller than in the no retrospective rebate model, reflecting a higher likelihood of families seeking out-of-pocket services under integrated care. This is because the integrated system aims to alleviate pressure on the AFP by encouraging more families to seek services from the private sector.

The plots in figure 6 further confirm the results established in corollaries 1 and 4, with the exception of the trend observed for c_w . Panel (a) shows that v_T^* increases with c_w , as higher queue management costs incentivize the AFP to expand service capacity, making it more appealing for families to wait for the AFP funds. This discrepancy arises because, in figure 6, μ^* is incorporated into the function $v_T(\mu)$, whereas in the corollaries, μ is treated as a fixed parameter in deriving monotonicity properties of delay sensitivity threshold.

Overall, the analysis in this section shows that the service capacity tends to be higher in the integrated care system. Offering rebate payments brings the system closer to the integrated care model from its current state. However, rebates encourage more families to seek out-of-pocket services, which could be greater or lesser than what the integrated system can achieve: as seen in figure 6, the value of v_T^* in the integrated model intersects with that of the retrospective rebate model. Thus, in the following section, we explore the implications of these findings for the various stakeholders.

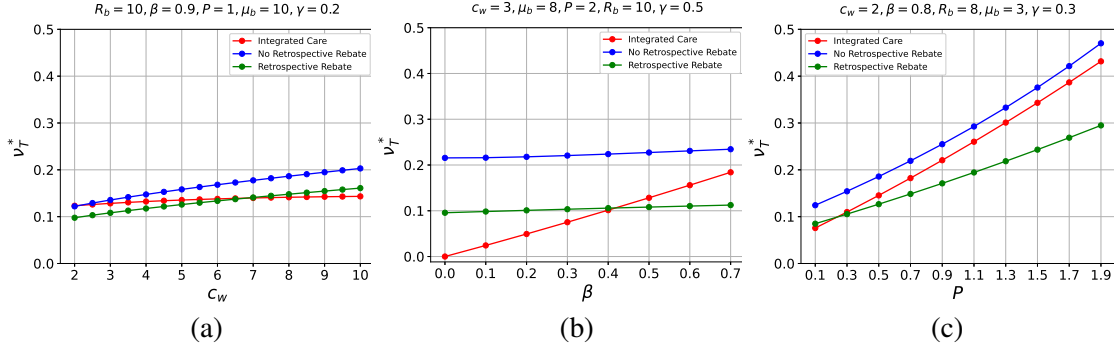


Figure 6 Optimal delay sensitivity threshold, v_T^* , as a function of different model parameters.

7 Implications to Stakeholders

In this section, we evaluate families' total utility denoted by TU , AFP's expected payoff V^* , overall out-of-pocket money spent denoted by M , and expected waiting time w_s for each system using the corresponding optimal service capacity and delay sensitivity threshold. The families total utility is calculated by summing the utility of all families. In particular,

$$TU = \lambda \int_0^{v_T(\mu^*)} \left(\frac{R(\mu^*)}{1 + v w_s(\lambda, \mu^*)} \right) dv + \lambda \int_{v_T(\mu^*)}^1 (R(\mu^*) - (1 - \gamma)(H + P w_s(\lambda, \mu^*))) dv. \quad (7)$$

The total out-of-pocket money spent is also obtained as

$$M = \lambda \int_{v_T(\mu^*)}^1 (1 - \gamma)(H + P w_s(\lambda, \mu^*)) dv, \quad (8)$$

which gives the total number of families seeking out-of-pocket services multiplied by the overall out-of-pocket expenses including base fee and the variable price per unit of time spent in the system—adjusted for the rebate. The AFP's expected payoff is evaluated using the objective function in problem (3). Remember that in the above calculations, we assume $\gamma = 0$ for both the integrated care system and the no retrospective rebate model.

The plots in figure 7 illustrate the expected waiting time as a function of various model parameters for the three systems. As expected, the integrated care system consistently yields the lowest waiting time, serving as a benchmark. Panel (a) shows that for approximately $c_w \geq 6$, all configurations perform similarly. Panel (b) demonstrates that offering rebates reduces the expected waiting time by approximately half across all values of β . Panel (c) illustrates that the gap between the integrated care system and other models increases with out-of-pocket expenses P . Finally, panel (d) shows that rebate rate γ has little effect on expected waiting time when below 20%, but becomes significantly impactful at higher levels, particularly between 80% and 90%. These results suggest that introducing rebates helps reduce expected waiting time by incentivizing higher service capacity, thus aligning system performance more closely with the benchmark.

The plots in figure 8 investigate the impact of rebates on different stakeholder's outcomes. Panel (a) shows that the integrated care system results in highest level of social welfare and that increasing rebates improves

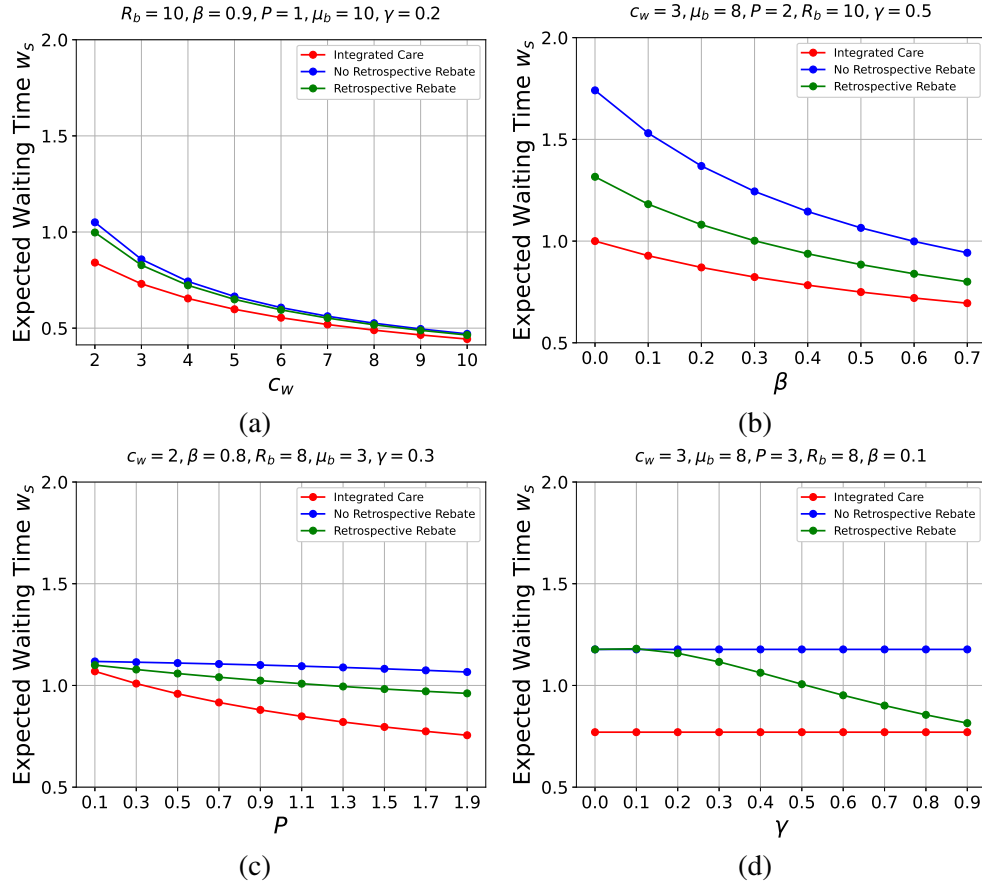


Figure 7 Expected waiting time as a function of different parameters for each care system

overall social welfare, bringing it closer to the central planner's objective. As shown in panels (b) and (c), offering rebates is likely to reduce total out-of-pocket spending and increase total utility, respectively. However, this strategy may negatively affect the AFP by reducing its expected payoff, as shown in panel (d).

Overall, rebates improve social welfare and benefit families through reduced out-of-pocket spending and higher utility, but they impose a trade-off by reducing the AFP's expected payoff. To further examine the impact of rebates on each stakeholder's outcomes, we analyze the optimal rebate level that benefits both parties, highlighting the inherent trade-off. In the model with retrospective rebates, the families' total utility is evaluated as defined in equation (7), while the AFP's expected payoff is determined using the objective function in problem (3). To facilitate comparison, we apply min-max normalization to both outcomes. In particular, we identify the minimum and maximum values in each outcome and map them to zero and one, respectively. All other values are then linearly scaled within this range, preserving their relative positions. Figure 9 presents the normalized outcomes of families and the AFP, as well as their combined total, as a function of rebate rate γ . The results suggest that when the two objectives are equally weighted, a balance between families' total utility and the AFP's expected payoff is achieved at approximately $\gamma \approx 0.57$ in the illustrated case, where maximum benefit is generated from introducing rebates.

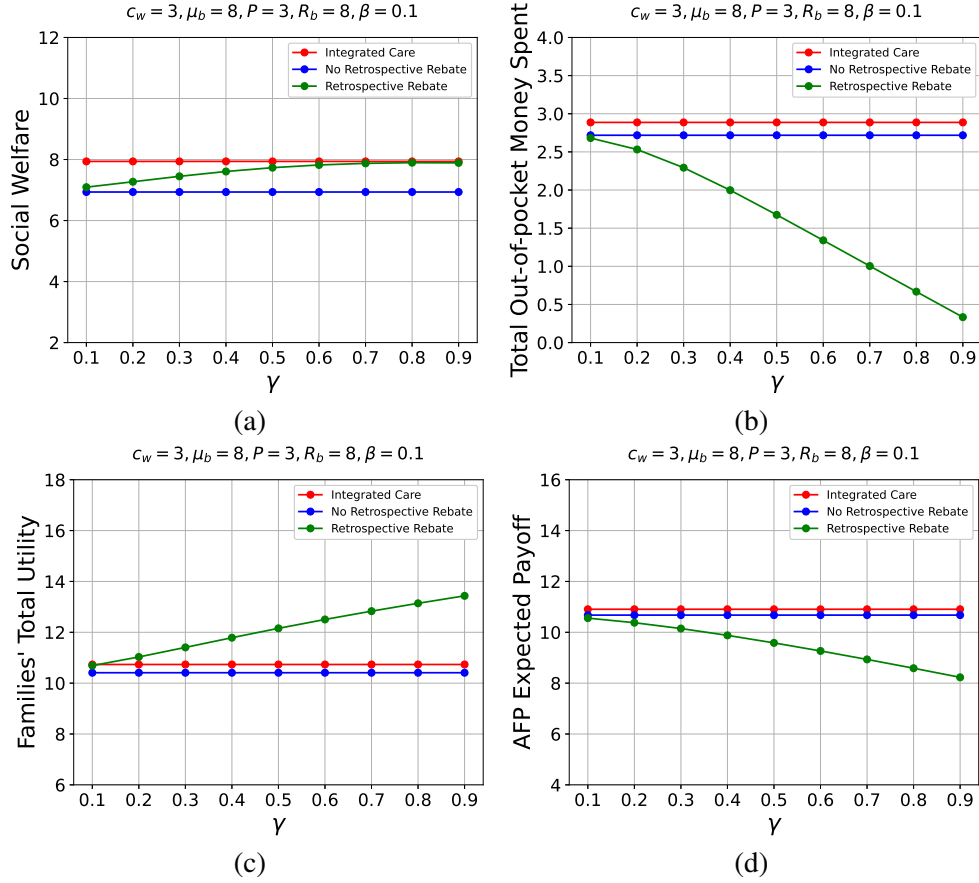


Figure 8 Different stakeholders' outcomes as a function of rebate rate γ

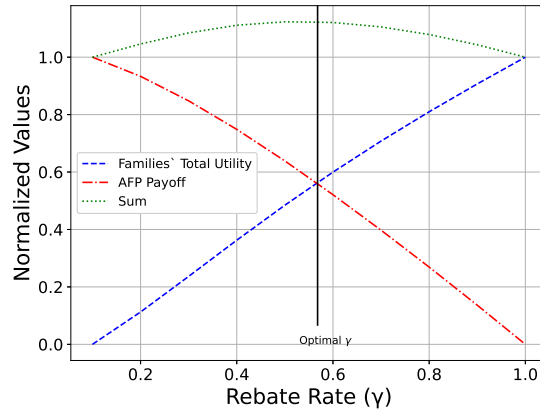


Figure 9 Optimal rebate rate considering families' total utility and the AFP's expected payoff with $c_w = 3$, $\beta = 0.1$, $P = 3$, $\mu_b = 8$, and $R_b = 8$.

Panel (a) of figure 10 illustrates the difference between the normalized families' total utility and the AFP's expected payoff as a function of γ for various values of c_w . It implies that there exists a threshold for γ : below this threshold, the AFP's loss from providing retrospective rebates outweighs the corresponding gains in families' total utility; above it, the gain in families' total utility exceeds the AFP's financial loss.

The figure further shows that this threshold—and hence the optimal rebate rate— increases with c_w . This is because with higher cost of managing the queue, the AFP becomes more willing to offer higher rebate rates in order to reduce queue length and avoid the high costs associated with congestion. Panel (b) presents the same analysis as panel (a), but varies β instead of c_w . In contrast to the trend observed for c_w , the figure shows that the optimal rebate rate decreases with β . This is because, as the cost of managing the queue for families who seek out-of-pocket services becomes comparable to that of those who wait exclusively, the incentive for the AFP to reimburse private expenses diminishes, thus we observe a reduction in the optimal rebate rate.

In summary, the analysis above reveals that the optimal rebate rate increases with c_w and decreases with β , reflecting how the AFP adjusts its willingness to subsidize private expenses in response to queue management costs.

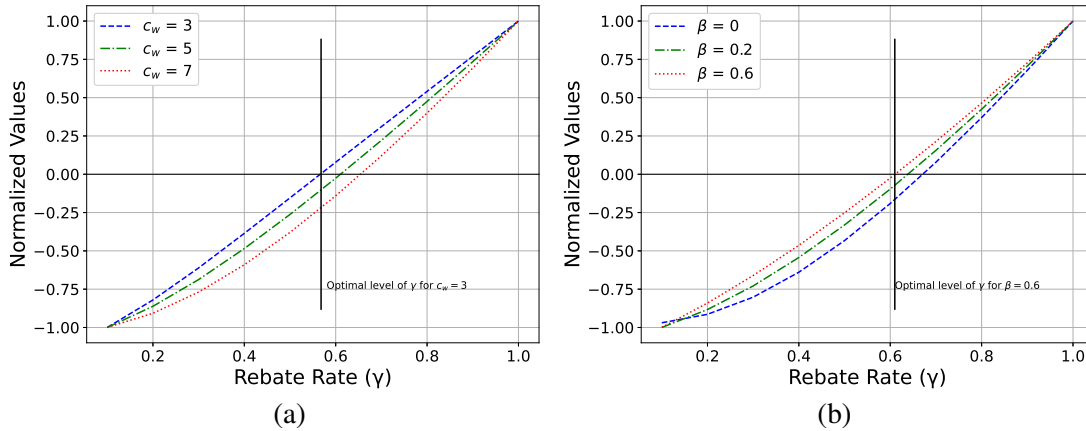


Figure 10 Difference between families' total utility and the AFP's expected payoff as a function of rebate rate γ ; Panel (a): for different values of c_w with $\beta = 0.1$; Panel (b): for different values of β with $c_w = 7$. Illustrated for $P = 3$, $\mu_b = 8$, and $R_b = 8$.

In what follows, we study the sensitivity of different stakeholders outcomes to various model parameters. When applicable, the analysis incorporates the corresponding optimal rebate rate as obtained and explained in figure 9. The top panels in figure 11 shows these outcomes as a function of variable out-of-pocket price P . A family's total utility decreases with P , reflecting the financial burden on families as costs rise (see figure 11, panel (a)). Panel (b) reveals a concave relationship between total out-of-pocket money spent and P . For low values of P , total spending increases as P rises, which is expected due to higher per-family expenses. However, for higher values of P , spending begins to decrease as the high costs deter families from seeking out-of-pocket services, reducing overall out-of-pocket spending. As shown in panel (c), the AFP's expected payoff remains relatively insensitive to changes in P .

The bottom panels in figure 11 shows similar stakeholder's outcomes as a function of AFP out-of-pocket care factor β . Panel (d) shows that families' total utility remains relatively unaffected by changes in β . Panel (e) reveal that the total money spent by families seeking out-of-pocket services decrease as β increases for

the no retrospective rebate model and the integrated care system. This trend can be due to the increase in the optimal delay sensitivity threshold v_T^* when β increases, as established in section 6. The increase in v_T^* means fewer families opting for out-of-pocket services, thereby reducing total expenditures. This panel further shows that total out-of-pocket money increases with β in the retrospective rebate model. This is because optimal rebate rate decreases with β as shown in figure 10, panel (b). Panel (f) of figure 11 demonstrates that the AFP's expected payoff decreases with increasing β . This decline occurs because a higher β raises the queue management cost for families who opt for out-of-pocket services, thereby reducing the AFP's net profit.

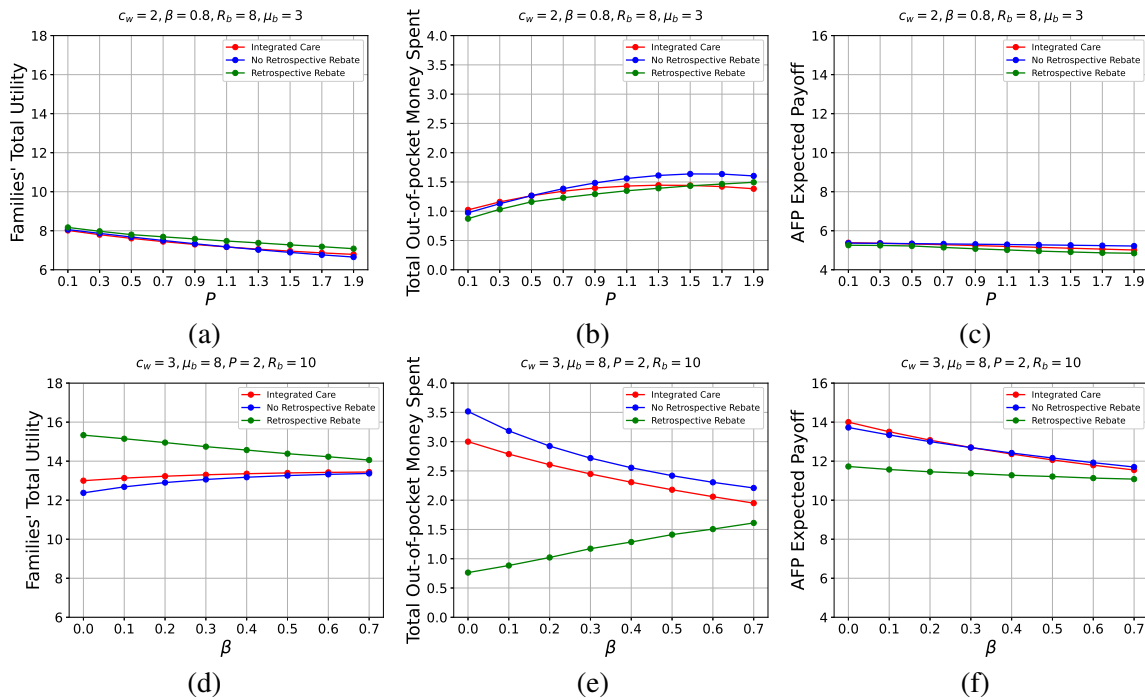


Figure 11 Different stakeholders' outcomes as a function of variable out-of-pocket price P (top panel), and the AFP out-of-pocket care factor β (bottom panel).

To better understand the conditions under which the integrated care system achieves better outcomes for families and the AFP compared to the no retrospective rebate model, Figure 12 presents a comparison of outcomes between the two models as functions of various parameters. Panel (a) shows that the integrated care system yields a higher total utility when the out-of-pocket price P is high and c_w is low to moderate. Panel (b) reveals a decision boundary in the (P, β) space, above which families spend more out-of-pocket under the no retrospective rebate model. Panel (c) demonstrates that the decision boundary for the AFP's payoff is nonlinear; notably, the no retrospective rebate model tends to yield a higher payoff for the AFP when β is high.

Overall, figures 11 and 12 show that the integrated care system does not always yield better outcomes in terms of families' total utility, total out-of-pocket spending, or the AFP's expected payoff. This is because

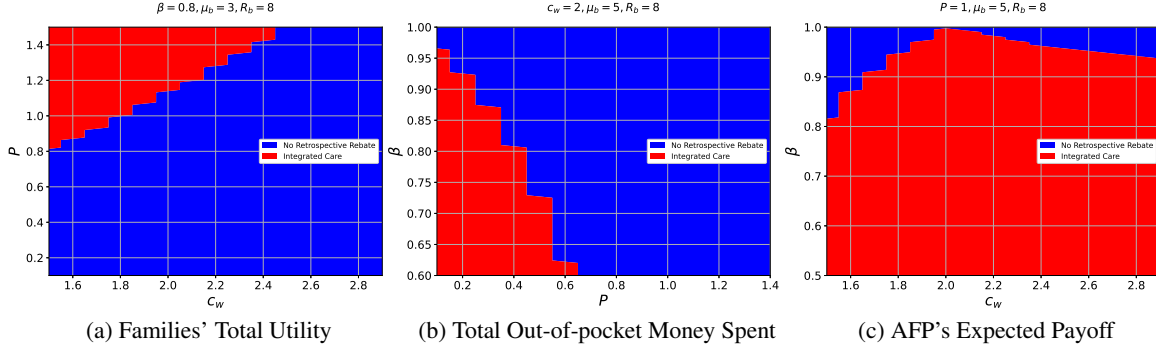


Figure 12 Comparison of the integrated care system and the no retrospective rebate model across different stakeholder

outcomes. The blue region indicates parameter combinations where the no retrospective rebate model yields higher outcomes than the integrated care system, while the red region represents the opposite. The boundary between the two regions marks the threshold where both systems perform equally.

the central planner's objective defined in problem (5) deviates from the typical social planner model found in the literature, which is often formulated as the sum of both parties' objectives. Instead, it improves upon the AFP's myopic decision-making by incorporating the families' discounted rewards when waiting exclusively for AFP funds. Therefore, policy makers can benefit from introducing a balanced rebate mechanism to close the gap between the current practices under no retrospective rebate model and the benchmark integrated care system.

8 Conclusions & Managerial Implications

We develop a queueing game framework that models the interaction between families seeking access to funds for autism services and the AFP responsible for assessments and funding allocation. Using an $M/M/1$ queueing model to represent service delivery dynamics, the AFP balances service capacity with financial constraints, while families decide whether to wait exclusively for AFP funds or seek private care based on the trade-off between timely intervention benefits and out-of-pocket service expenses.

In the retrospective rebate model, we show that families' decision follows a threshold policy based on their sensitivity to delays and characterize how service capacity influences their decisions. For the no retrospective rebate model, we analytically derive the AFP's optimal service rate and families' corresponding decisions. In particular, when the optimal service capacity is extremely low but stable, families are left with no choice but to wait exclusively for AFP funds, as private care becomes prohibitively expensive. At the highest optimal service capacity, families benefit the most, as they can rely on AFP funds without incurring private costs. For intermediate optimal service capacity levels, our findings suggest that families with higher sensitivity to delays would be better served if they had access to out-of-pocket care.

We further extend our framework to an integrated care system, where a central planner optimizes both service capacity and families' decisions to maximize social welfare. We characterize the optimal policies and our results show that this system achieves the highest service capacity and lowest waiting times, serving

as a benchmark. Retrospective rebates help bridge the gap between integrated care and current no rebate practices by reducing waiting times but they come at a financial cost to the AFP. To address this trade-off, we explore optimal rebate strategies that balance families' benefits with the AFP's financial sustainability. While the integrated care system serves as a benchmark for its shorter waiting times, it can lead to lower total utility for families. This is because it accounts for the timing of reward realization, which is discounted for those who wait exclusively for the AFP funding, aligning it more closely with families' actual expectations.

Given the critical role of timely intervention in improving long-term outcomes, our findings provide policy insights for designing incentive mechanisms that enhance access to autism services. This could involve offering balanced retrospective rebates to families, helping to partially cover their expenses during the waiting time. While increasing the service capacity and reducing waiting times may appear beneficial for families (as in the integrated care system), their total utility may be adversely affected due to the timing and structure of reward realization. This emphasizes the importance of selecting an appropriate rebate rate to align incentives.

Future research could explore alternative rebate structures, such as providing rebates only to families exceeding a wait-time threshold, and examine the effects of information asymmetry by relaxing the assumption that families have full knowledge of expected waiting times before making decisions. Additionally, extending the model to include funding allocation decisions, mismatches between allocated funds and children's actual needs, and competition among healthcare providers could yield deeper insights into system dynamics and policy design. Another direction is to relax the all-or-none assumption regarding families' service-seeking behavior by allowing for temporary or partial out-of-pocket engagement (e.g., fewer hours per week), which would adjust their choices based on evolving financial capacity.

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Appendix. Summary of Notations

Symbol	Description
Parameters	
P	variable out-of-pocket expense for every unit of time spent in the system
H	base fee for receiving out-of-pocket services
λ	arrival rate of families to the system
c_w	cost of managing the queue per unit of time patients spend in the system
β	the AFP out-of-pocket care factor,
γ	rebate rate
α	measures the sensitivity of the quality-driven reward to the service rate
μ_b	baseline service rate
R_b	baseline reward collected by the families for timely intervention
κ	coefficient of the AFP's operating cost
l_1	lower bound of μ which ensure v_T remains non-negative
l_2	upper bound of μ which ensure v_T does not exceed one
Decision Variables	
μ	the AFP's service rate
v_T	families delay sensitivity threshold
Random Variables	
v	delay sensitivity level of families with pdf $U[0, 1]$
Functions	
$w_s(\lambda, \mu)$	the expected waiting time spent in an $M/M/1$ queueing system with arrival rate λ and service rate μ with $\mu > \lambda$
$R(\mu)$	lifetime quality-driven reward for timely intervention collected by families as a function of the AFP service rate μ in monetary values
$C(\mu)$	the AFP operating cost function
$\phi(\mu)$	function given in equation (EC.5)
$\psi(\mu)$	function given in equation (EC.11)